

CHAPTER 3

Investigation and site selection for small hydropower system

3.1 Introduction

Investigations covering data collection and/or measurements are important for project planning, as well as designing. In comparison with large hydropower projects, investigation of small hydropower projects (SHP) needs to be simplified, fast and economical, and commensurate with the size of the components. The emphasis has to be that the methodology adopted must relate to the size and function of the project. Investigations are aimed at the collection of data in order to plan the proper location of components and optimum capacity of the project. Investigations need investment in terms of time and money. Thus the extent of data to be collected/measured depend on the magnitude of the project and requirement. During investigation, the following need to be carefully addressed:

1. Proper assessment of water flow as the power generating capacity can become abnormally higher or lower due to incorrect flow data, which is not desirable.
2. Proper assessment of high flood discharge, which is required for planning and designing of diversion structure and powerhouse levels.
3. Proper assessment of geological conditions for the safety of civil works structures.

3.2 Objective of site investigation

The main objectives of the site's investigation include:

1. Assessment of site suitability for various components.
2. Providing acceptable and cost-effective design for the components.
3. Solution to technical and nontechnical problems that may occur during all the stages of development.

In order to achieve the aforesaid objectives, there are several types of investigations required to be carried out for the schemes in hills and plains. The investigations for schemes on existing facilities are rather less published; otherwise, most of the data are available from the owners of these facilities. Following are the field surveys required to establish the SHP project schemes.

1. Topographical survey
2. Hydrological survey
3. Geological investigation

4. Load demand and power evacuation survey
5. Socioeconomic survey
6. Construction material survey
7. Environmental survey
8. Infrastructural survey
9. Muck disposal survey

3.3 Preliminary investigation

Before undertaking reconnaissance, a table study is undertaken based on available data on topographical maps, discharge, and geological studies. From this study, a rough scope of the project investigations and tentative location of various components are indicated. Thereafter, field reconnaissance for preliminary investigation is undertaken to have a general appreciation of the terrain and water flow to assess the feasibility of the project. The basic aim of the preliminary survey is to have knowledge regarding the characteristic of the site, to identify the difficulties and their solutions, and general layout for site selection. Before the preliminary visit of a site, data available should be studied and initial designs prepared to conceptualize them on the map of the site during the preliminary survey. Further, it is also suggested to prepare an outline of the reconnaissance description along with a checklist. It will assist in recognizing significant information to be investigated/surveyed during the site survey.

The purpose of the preliminary investigations is to provide sufficient information to confirm the practicability of any proposed scheme of development and to compare alternative schemes or arrangements so that preliminary design and estimates can be prepared and recommended with reasonable confidence. These preliminary investigations generally include the following studies:

1. Collection of hydraulic data (precipitation, run off, flow in the river, etc.) from the available records. In case past discharge data are not available, discharge is measured once or twice by using simple methods with available instruments such as a flow meter or by float method when flow meters are not available. The assessment of discharge in different seasons is made by a few measurements and local inquiry on possible water levels in different seasons. The discharge data are assessed for the whole year for planning the installed capacity of the project.
2. A preliminary survey is carried out by using available instruments such as an altimeter, Global Positioning System, engineering level, theodolite, plane table, and total station. The available topographical maps can also be used for preliminary planning of the project. These maps may be available in the scale of 1:50,000 or 1:25,000, which may not give accurate planning for smaller areas.

3. Overall geological features, local characteristics, and some surface surveys are carried out for assessment of the geology of the project for planning the layout of various components.
4. Assessment of power demand scenario in the nearby area of the project, as well as determination of infrastructure available for power evacuation are also carried out on the basis of local inquiries. Available census data are required to plan the capacity of the project in an isolated scheme. For grid-connected SHP projects, such surveys are not required.

With such investigations, a reconnaissance report will be prepared and discussed for taking further decisions on whether detailed investigations are required or not. Such a study is designed to reduce the chances of a subsequent failure. In case the project is found nonfeasible (technically), it may be dropped at this stage. For technically feasible projects, detailed surveys and investigations are carried out. The scope and extent of the site investigations depend partly on the availability and reliability of existing published data, the nature and size of the proposed development, and opportunities/difficulties presented by the existing topographical and geological features, which in turn, will affect the number of alternatives that have to be carried out. For these reasons, the various classes of site investigations cannot readily be tabulated in any invariable order or priority or importance. SHP projects are broadly two types (given below) and the investigations will be different for both types.

- Projects in hills (basically run-off river type) using the steep slopes of streams. These are of medium and high head.
- Projects on existing facilities such as irrigation dam and canal fall structure. These are of low or medium head.

While there are several types of investigations required to be carried out for the schemes in hills, the investigations for schemes on existing facilities are rather fewer. The final investigations comprise the detailed exploration of the finally recommended site (or sites) so as to establish their complete suitability beyond doubt and to enable the final designs and contract documents to be prepared with the least chance of serious alterations after the contract has been let.

3.4 Detailed investigations

The detailed survey and investigations are carried out with the objective of preparing a topographical map of the area on a suitable scale, establishing ground flow measurements, and other surveys as detailed below.

3.4.1 Topographical survey and site selection

After carrying out the survey and preparing a preliminary report, the advantage of topographical investigation is to map the positions of the main components, which have almost been finalized. Generally, the topographical representation (mapping) should be carried out in such a manner that it sufficiently covers all alternatives visualized during the field visit, as well as allows the possible realignment of civil structures (such as waterways and access roads) and routes of the electricity transmission line. The complete network of hydropower schemes comprised many components from intake structure to tailrace system, including switchyards area, laydown areas, and the accessed road of the site.

An investigation is required on the upstream cross-section of the diversion structure (barrage/dam) in order to assess the maximum head limit, area of stored water in the reservoir, and its volume. The survey should be carried out on a number of cross-sections with specified spacing. Spacing between two consecutive cross sections varies from 2 to 5 times the river width, but it should be lesser than 500 m. The current level of water and highest watermarks need to be determined. The surveyed sections should be specified as a reference central line. If the flow of water is high or deep, then the measurement of water depth is determined either visually or by using a sounding instrument. Along the central line of the dam, three to five cross sections need to be surveyed. Surface mapping is significant to include abutments of the barrage, as well as head regulator area.

In order to create a time plot of available discharge/head, survey of the river's cross-section is required at the downstream section of the diversion dam. The survey/investigation continues to some distance on the downstream section for determining the drop requirement between the rivers (on just the downstream section) and the desilting device.

As far as possible, topographic maps from various agencies, such as the Survey of India and other state agencies, should be collected first. In India, almost the entire region has a scale map with a 1:50,000 contour interval of 40 m and a 1:25,000 contour interval of 10 m. Lesser surveys/investigations are required for low-head schemes.

- In the case of existing SHP dams and diversion structures, no investigation may be required where penstock pipes are already embedded.
- In case the penstock is not embedded, a separate water-conductor system will be required to carry the water to the location of the powerhouse.
- If the power plant is built in an existing channel where there is sufficient water head, a bypass mechanism is required. The survey will be conducted in a short period of time to determine the best channels/detours, etc.

Contour plans of the proposed arrangement should be prepared on a scale of 1:500 to 1:1000 with a contour spacing of 1–2 m during field surveys. The L-section and channel cross-section must also be prepared.

Civil works for the SHP plan include a diversion weir and intake structure, a power channel, a desilting tank, a forebay, a spillway, a penstock, a surge tank, a powerhouse, and a tail race channel. In all types of schemes, all of these or lesser components may be there. Thus, site selection in different types of schemes is different, as detailed below.

3.4.1.1 Canal-based small hydropower schemes

In such schemes, the water flowing in the canal and available head on the canal fall are used for power generation. Thus powerhouse needs to be located in the vicinity of the canal fall. Such schemes have diversion channel, powerhouse, and tailrace channel as civil works components. The location of the powerhouse can be on the right or left side of the canal or in the canal itself, where spillway and powerhouse are cited together. Fig. 3.1 shows topographical surveys required for canal-based schemes. The following need to be taken into consideration for locating the diversion channel and powerhouse.

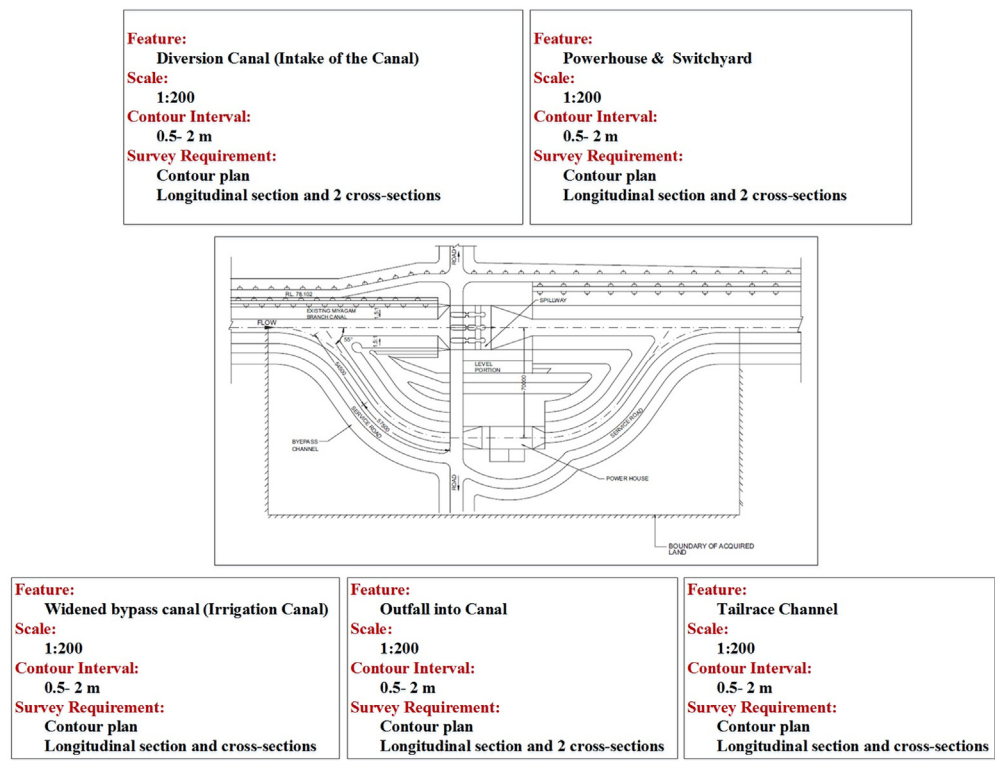


Figure 3.1 Topographical survey for canal-based scheme. From S.K. Singal, R.P. Saini, *Analytical approach for development of correlations for cost of canal-based SHP schemes*, *Renew. Energy* 33 (12) (2008) 2549–2558, ISSN 0960-1481.

1. Proximity of access road
2. Availability of land
3. Location of substation for power evacuation

3.4.1.2 Dam-Toe schemes

Dam-toe schemes are those where penstock is taken through the body of the dam and powerhouse is cited at the toe of the dam. Intake, penstock, and powerhouse are the main components in such schemes. Fig. 3.2 shows topographical surveys required for dam-based schemes. For locating these structures, there is limited choice, and the components need to be sited near the dam considering the following:

1. Geology of the area
2. Access road
3. Availability of land
4. Location of grid substation for power evacuation

3.4.1.3 Run-of-river schemes

In such schemes, water is diverted from the river through diversion weir and flows toward the forebay through the power channel. In the power channel, a desilting tank is provided to remove silt particles. From forebay to powerhouse, water is conveyed through a pressurized pipe known as penstock. Fig. 3.3 shows topographical surveys required for run-of-the-river schemes. Each component needs to be sited carefully as per following criteria.

3.4.1.3.1 Diversion weir and intake

The diversion weir is located in the river itself. Thus width of river should be as minimum as possible. Normally a river is not straight and will have many bends along its course. Bends should be avoided for locating the diversion weir. The area should be free from landslides, and the banks of the river should be stable for safe connectivity of the diversion weir.

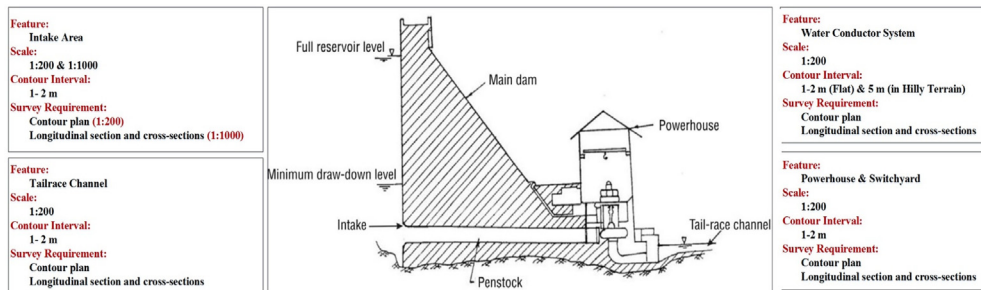


Figure 3.2 Topographical survey for dam-based scheme. From S.K. Singal, R.P. Saini, *Cost analysis of low-head dam-toe small hydropower plants based on number of generating units*, Energy Sustain. Dev. 12 (3) (2008) 55–60, ISSN 0973-0826.

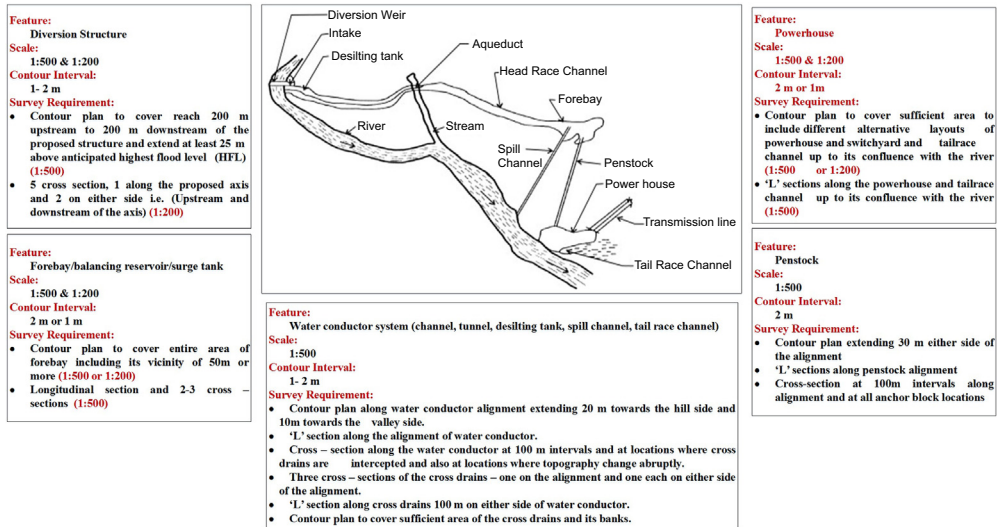


Figure 3.3 Topographical survey for run-of-river scheme. From S.K. Singal, R.P. Saini, C.S. Raghuvanshi, *Analysis for cost estimation of low head run-of-river small hydropower schemes*, *Energy Sustain. Dev.* 14 (2) (2010) 117–126, ISSN 0973-0826.

3.4.1.3.2 Power channel, desilting tank, and forebay

The location of these components should be on mild slopes in hilly areas to avoid excessive slope cutting. Also filled earth should be avoided. The alignment of the power channel may cross some drains. On each drain, cross drainage works are required to be constructed. Thus alignment of power channel should be such that it crosses minimum number of drains.

3.4.1.3.3 Penstock

Penstock connects forebay and powerhouse. The alignment of penstock should be such that it has (1) minimum length and (2) minimum bend (horizontal, as well as vertical), and the alignment also should preferably be along the ridge and not through the valley.

3.4.1.3.4 Powerhouse

The powerhouse should be located on fairly level ground in a landslide-free area. It should be located well above the high flood level of a nearby river.

3.4.2 Hydrological investigation

A hydrological survey is carried out to assess the amount of water available for power generation. Accurate measurement of river flow, ideally at the actual location to be

intercepted, is the most reliable technique for estimating the flow. The approaches available for solving hydrological challenges are:

1. Existence of a lengthy sequence of observations of flow data is a very unusual occurrence for small rivers.
2. The flow data with a short sequence of field observations is supplemented by indirect data associated with the gauged station.
3. The flow data available at the location, as well as the data from the reference station, are not accessible.

In the third approach mentioned above, five primary approaches to the answer are commonly used: hydrological analogies and investigation, correlation analysis, regional synthesis, or any combination of the aforementioned approaches. There may be little data available for determining reliable flow in run-of-river systems, particularly for SHP projects. In such circumstances, discharge observations should begin as soon as possible at a suitable location. In order to get the most trustworthy correlation, the data obtained on the basis of recorded rainfall should be analyzed to derive a long-term run-off series. In order to determine a correct connection between rainfall and run-off, the rain gauge station should be placed within the catchment area, or in the absence of such a site, the rain gauge stations should be placed in nearby catchments. Hydrological investigation involves the collection of rainfall/snowfall data, climatological data (temperature, humidity, and wind), gauged river discharges, sediment load, and water quality-related data. There are a number of direct and indirect methods for measuring discharges, and any of the following (each with its own set of benefits and drawbacks) may be used, depending on the circumstances. [Table 3.1](#) provides the details regarding the discharge measuring instruments.

- Direct methods: Velocity area approach, dilution technique using constant-rate injection or sudden injection method, ultrasonic and electromagnetic technique.
- Indirect methods: Flow measuring notches, weirs, and flumes, slope-area, and stage-discharge approach.

The discharge measurements should be taken over at least a two-year period and following are the important properties that will be examined in addition to discharge.

1. Sediment in suspension
2. Water levels in the tail race
3. Daily and average dissolved oxygen
4. Water temperature, average and daily

3.4.3 Geological investigation

Geological investigation involves the preparation of a geological map of the project site on the basis of surface observations along with laboratory tests of collected samples. As far as SHP projects are concerned, maximum judgment is based on a visual investigation of site features to assess the geologic condition of the site. Faults, fractures, shear

Table 3.1 Details regarding the discharge measuring instruments.

Name of the instrument	Working principle
Gauges (staff, needle, wire-weight, float)	Gauges are several sorts of devices that provide water level readings (in meter). This device is a graded scale that is fixed on the side bank in the water and can be read by looking at the water level surface. The water level gauge should be situated such that it can be easily read and recorded from either bank.
Recorders	These are a collection of devices that create a time-record of water surface elevation. Recorders can produce analog or digital output.
Floats	Any buoyant object can be used as a float. It must be bulky enough to extend below the water line to a suitable depth. Because a float measures surface velocity, an appropriate correction factor is used to calculate mean velocity (0.8–0.9).
Current meters	There are two types of current meters: • <i>Cup-type instrument</i> has a vertical axis, contains vanes, a rod to hold the meter, heavy weight to sink the meter when hanging on a cable, and an electric device to receive/emit signals to count the rpm. • A <i>propeller current meter</i> has a horizontal axis. When compared to cup-type meter, it offers a few benefits. It is lesser affected by the components of velocity that are not parallel to the meter axis. They are smaller and more suited to multiple unit installation.
Electromagnetic current meter	Electromagnetic current meter produces voltage and it is relative to velocity. These devices have considerably improved in recent years; however, they are still not as reliable as anemometers.
Ultrasonic (acoustic) velocity meter	Acoustic flow meters are nonintrusive, nonmechanical devices that can measure discharge in pipelines or open canals. This device can record flow rates continuously and reliably under a variety of circumstances, including flow in both directions.

zones, type of rock, strata, huge cracks (open), fissures, position of springs, signs of erosion, landslides, rock falls, and rock bursts, are some of the major characteristics that indicate the geological condition of the site. The condition of the foundation for various structures is assessed on the basis of the obtained data from the survey/investigations, as well as from laboratory tests. Figs. 3.4–3.6 provide details regarding the geological investigation of various components of canal-based, dam-based, and run-of-river schemes, respectively.

Structure: Water Conductor System No. of Trial Pits: 4 Location of Trial Pits: (1) At inlet – 1 (2) U/s of powerhouse - 1 (3) D/s of powerhouse – 1 (4) At outlet – 1 Depth of Trial Pits: 1.5- 2 m Additional Requirement: Geological assessment of stability of excavation slopes	Structure: Powerhouse No. of Trial Pits: 1 Location of Trial Pits: Powerhouse Area Depth of Trial Pits: 1.5- 2 m Additional Requirement: Geological assessment of excavation slopes and powerhouse foundation grade
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Figure 3.4 Geological surveys for canal-based schemes. From AHEC-IITR, “1.12 – Site investigations” standard/manual/guideline with support from Ministry of New and Renewable Energy, Roorkee, August 2013.

Structure: Intake No. of Trial Pits: 1 Location of Trial Pits: Intake Area Depth of Trial Pits: 1.5- 2 m Additional Requirement: Geological assessment of stability of excavation slopes	Structure: Water Conductor System No. of Trial Pits: 3 Trial pits (Open Channel) 3 Drill hole (Tunnel) Location of Trial Pits: Along alignment (Open Channel) Intake portal, Outlet portal and Low cover reach (Tunnel) Depth of Trial Pits: 1.5- 2 m (Open channel) Delineation of shear zones up to tunnel grade (Tunnel) Additional Requirement: Geological assessment of stability of excavation slopes (Open) Geological assessment along alignment (Tunnel)
Structure: Powerhouse No. of Trial Pits: 1 Location of Trial Pits: Powerhouse Area Depth of Trial Pits: 1.5- 2 m Additional Requirement: Geological assessment of stability of excavation slopes and powerhouse foundation grade	Structure: Tail race channel or Tunnel No. of Trial Pits: 3 Trial pits (Open Channel) 3 Drill hole (Tunnel) Location of Trial Pits: Along alignment (Open Channel) At portal, along alignment and at low cover reach (Tunnel) Depth of Trial Pits: 1.5- 2 m (Open channel) Up to tunnel grade (Tunnel) Additional Requirement: Geological assessment of excavation slopes

Figure 3.5 Geological surveys for dam-based schemes. From AHEC-IITR, “1.12 – Site investigations” standard/manual/guideline with support from Ministry of New and Renewable Energy, Roorkee, August 2013.

Structure: Diversion Structure No. of Trial Pits: 3 Trial Pits (Conventional Weir/Trench Weir/Barrage) 3 Drill hole (Low Dam) Location of Trial Pits: One trial pit at middle and two trial pits, on either abutment end (Conventional Weir/Trench Weir/Barrage) One borehole at middle and one borehole at intake location (Low Dam) Depth of Trial Pits: 1.5- 2 m (Conventional Weir/Trench Weir/Barrage) Up to Foundation level of Dam (Low Dam) Additional Requirement: Geological assessment of stability of rock slopes on either abutment	Structure: Water Conductor No. of Trial Pits: 3-5 Trial Pits (Open Channel) 3 Drill hole (Tunnel) Location of Trial Pits: Trial pits at every 500 m c/c and at critical locations (Open Channel) Intake portal, Outlet portal, Low cover reach (Tunnel) Depth of Trial Pits: 1.5- 2 m (Open Channel) Up to Tunnel Grade (Tunnel) Additional Requirement: Geological assessment of stability of rock slopes along the alignment reach-wise (Open Channel) Delineation of shear zone, Geological assessment along alignment, Geophysical assessment (Tunnel)		
Structure: Penstock No. of Trial Pits: 2 Location of Trial Pits: Along the alignment of penstock preferably at anchor block locations. Depth of Trial Pits: 1.5- 2 m Additional Requirement: Geological assessment of penstock slope	Structure: Forebay No. of Trial Pits: 2 Location of Trial Pits: One trial pit in forebay area and another trial pit at penstock intake location Depth of Trial Pits: 1.5- 2 m Additional Requirement: Geological assessment of rock slope on hill slide of forebay	Structure: Powerhouse No. of Trial Pits: 1 Location of Trial Pits: Powerhouse Area Depth of Trial Pits: 1.5- 2 m Additional Requirement: Geological assessment of excavation slope	Structure: Tail race channel No. of Trial Pits: 1 Location of Trial Pits: Along alignment Depth of Trial Pits: 1.5- 2 m Additional Requirement: Geological assessment of excavation slope

Figure 3.6 Geological surveys for run-of-river schemes. From AHEC-IITR, "1.12 — Site investigations" standard/manual/guideline with support from Ministry of New and Renewable Energy, Roorkee, August 2013.

3.4.4 Load demand survey

If the scheme is in isolated mode, a load demand survey is necessary to determine the area's power consumption. There are two types of schemes, and the investigations required are as follows:

3.4.4.1 Grid connected

The studies will include information on the nearest substation, its capacity, the length of transmission line necessary, the requirement of land, etc. The following surveys will be conducted:

1. The current state of the region's power supply, as well as system loads. Major loads to be supplied and energy requirements.
2. A diagram depicting the transmission line path for power evacuation.
3. A study of a nearby grid substation where power is anticipated to be supplied is being conducted. The research will look at the substation and transmission line carrying capacity, as well as the possibility of expansion due to the increased power.

3.4.4.2 Isolated schemes

Domestic/community/institutional/commercial/industrial load requirements need to be determined on a daily, seasonal, and annual basis. The size and quantity of units will be determined by the load curve. During the electrical load survey, the following elements should be taken into account to assess the electricity demand.

1. The total number of villages.
2. The number of residences that are occupied.

3. Population.
4. The total number of anticipated connections.
5. Daily energy usage.
6. Street lighting needs.
7. The number of commercial establishments (stores, businesses) and the energy consumption for each one.
8. Energy usage from a number of health clinics, schools, and other community services.
9. The number of small industries and their energy requirements.
10. Unprecedented demand.
11. Current and forecast demand for various forms of electrical energy usage.

3.4.5 Socioeconomic investigations

For estimating the installed capacity of SHP, investigations of power users are critical. An SHP project's design load is anticipated in 5 years after the plant's commissioning, implying that at such a load level, the facility will be fully operational. The survey on socioeconomic aspects helps in the assessment of growth in power demand with time. Such surveys are not required on grid-connected schemes.

3.4.6 Meteorological surveys

Meteorological information is normally collected or available from the following sources such as temperature, rainfall, humidity, and wind velocity.

1. Meteorological stations located in the project area.
2. Regional information from meteorological departments

The analysis is carried out to assess the flow data based on catchment area and meteorological data for large projects or ungauged catchments where measured data are not available.

3.4.7 Construction material survey

Large quantities of material would be required for construction of the project, while most of the materials such as cement and steel will be procured from the open market. Coarse and fine aggregate required for the concrete mix will have to be sourced from the project area, preferably from sites close to the work areas. During the construction of the project, a crushing plant can be established near the project area. A part of the excavated material from the channel and other components can be used as construction material. Suitability of the above sourced material needs to be confirmed through laboratory tests as given below:

1. Petrographical examination of rock samples.
2. Tests on aggregate samples for use in concrete.

- a. Specific gravity
- b. Water absorption
- c. Soundness
- d. Aggregate impact value
- e. Aggregate abrasion value
- f. Aggregate crushing value
- g. Alkali aggregate reaction
- h. Flakiness index
- i. Elongation index
- j. Fineness modulus
- k. Gradation analysis

These tests need to be conducted during the execution stage of the project.

3.4.8 Load survey and power evacuation

Before the power potential analysis, a load assessment for power demand is required for isolated/off-grid projects to ensure that local demand will be satisfied by the installed capacity, as well as to determine the need for power growth over the next 5–10 years. To achieve higher dependability for isolated projects, the installed capacity of the project should be equivalent to 90% or higher reliable discharge. Lower reliability may be used if perpetual water is not available, but for those projects, a fusion (hybrid) system using diesel, biomass, or other sources can be involved during design stage. The availability of grid and quality of the grid connection for power evacuation must be evaluated. An independent power producer might be a merchant plant, captive plant, or traditional plant that supplies power to a distribution licensee. As a result, after studying the power evacuation system plan, the load assessment/grid status must be assured.

3.4.9 Environmental investigations

SHP schemes have a minimal adverse impact on the environment; however, in some cases, some impact may be noticed. The investigation is required on following aspects as per the policy of the concerned government and regulation.

1. After the reservoir was built, the foundation and submergence had an impact.
2. Deforestation has a negative impact.
3. Scarring or defacing the natural environment, particularly around borrow pits and quarry sites.
4. Cultural and historical arts are preserved.
5. It has an impact on the flow of the river.
6. Water quality changes (organic matter content, temperature, dissolved solids contents, etc.)

Environment impact assessment needs to be prepared based on data collected, and environmental mitigation plan is prepared to plan mitigation of negative impacts.

3.4.10 Infrastructural facilities

The following infrastructure facilities will be surveyed in detail, and their positions will be noted on the plan map according to project requirements.

1. Access roads within the project and project area
2. Power lines for the supply of construction power
3. Telecommunication facilities necessary during and after the project's construction
4. Colonies/residential structures for project personnel
5. Workshop
6. Drainage surrounding the power plant in the workshop
7. Sanitation and sewage

3.4.11 Muck disposal

The amount of waste that will be generated during construction activities of the project needs to be estimated. In the project region, some amounts of recyclable waste will be used for activities like earth filling. Unused muck must be disposed off on barren terrain that can serve as a muck disposal place. Wherever necessary, the area of waste disposal might be maintained by developing barriers to prevent the entry of waste into the nearby water body. Size of waste disposal sites should be provided in the project's general layout design, and their usage must be agreed to by the state forest department and/or concerned authority.

3.5 Flow duration

A flow duration curve is drawn based on the measured/collected discharge data. Data from the last two to ten years should be used to plot a flow duration curve. In the case of SHPs, however, fewer years' data can be used depending on availability. Flow duration curve is the basic tool for the determination of installed capacity.

A flow duration curve indicates the percentage of time that the flow is equal to, less than, or higher. The same data may be displayed to illustrate how many times certain flows are equaled or exceeded in percentage of time. On a 10 daily or monthly basis, a duration curve can be created. The range of the curve diminishes as the length of the time unit rises. The choice of time unit is determined by the curve's purpose. If a proposal for diversion without storage is being considered, the time unit should be daily in order to show absolute minimum flows. Depending on reservoir size in proportion to flow, a month may be enough for reservoir design.

The flow duration curve's primary limitation as a design tool is that it does not display flow in a natural order of time. The flow duration curve's shape is mostly determined by the time unit used (daily, monthly, or annual) and the kind of stream (i.e., highly variable discharge, small flow variability, large flood plains, or considerable base flow).

3.6 Power equation

Mathematical representation of hydropower generation is provided below in Eq. (3.1):

$$P = 9.81 \times \eta \times Q \times H \quad (3.1)$$

where, η is the overall efficiency, Q represents flow discharge (cumec), H designates net head availability (m), P denotes hydropower potential (kW).

3.7 Power potential and installed capacity

The following conditions must be met prior to determining installed capacity:

1. Applicable efficiencies, including part load efficiency of turbine, generator, and gearbox.
2. Evaluation of the rated head.
3. According to regulations, an environmental flow (minimum discharge) must be discharged in the stream, just downstream of the diversion structure. This passed discharge will be deducted from the available water to assess the water available for power generation.
4. The water conductor system has been designed to account for silt flushing outputs and other losses.
5. Other water usage requirements, such as drinking, irrigation, industrial, navigational, recreational, and so on, are satisfied and subtracted from the available water as needed.

3.7.1 Fixation of installed capacity

If discharge data is only accessible for a limited period of time, say less than ten years. The processes for determining an SHP scheme's installed capacity are as follows.

1. For all hydrological years, obtain daily or 10-daily average flow (year-wise) in m^3/s .
2. To assess water available for electricity generation, subtract environmental flow.
3. Arrange the discharge statistics in descending order for each year.
4. Calculate the average discharge for all of the data from the year.
5. Calculate the percentage of time that average discharge values correspond to and draw the flow duration curve.
6. Using the power equation, calculate the power potential for each discharge value.

7. Consider a range of installed capacity options in appropriate increments, such as 100, 200, 500, 1000, 1500, 2000, 3000, 4000, ... kW.
8. Calculate incremental energy generation (kWh) for each incremental capacity (MW) and display the curve between installed capacity and corresponding yearly energy for a typical project, as shown in Table 3.2 and Fig. 3.7. The curve drops sharply to around 21 MW, as seen in Fig. 3.7, implying that the installed capacity is likely to be about 21 MW. The installed capacity is set to a number where the graph's decline is steep.
9. The installed capacity option that delivers more energy generation at a reasonable/saleable generating cost compared to alternate source costs is chosen.

A cash flow analysis for the life of the plant (which may be regarded as 40 years) should be conducted for different installed capacities of the plant to determine acceptable/saleable generating costs. The cash flow series that will provide a positive Net Present Value, a sufficient Internal Rate of Return as required by the developer/financial institutions, and the permissible generation cost (which will depend on the regulated tariff or the cost of purchasing power for captive use or the cost of sale for Merchant plant, as the case may be) will guide the installation of capacity.

Table 3.2 Installed capacity versus incremental energy.

S. No.	Inst Cap. (MW)	Annual energy (GWh)	Load factor (%)	Incremental benefit $\Delta MU/\Delta MW$
1.	5	42.45	96.91	7.51
2.	6	49.91	94.96	7.28
3.	7	57.03	93.01	7.02
4.	8	63.90	91.17	6.62
5.	9	70.61	89.56	6.35
6.	10	77.09	88.00	5.92
7.	11	83.08	86.22	5.71
8.	12	88.82	84.50	5.34
9.	13	94.08	82.62	5.19
10.	14	99.18	80.87	4.91
11.	15	103.88	79.06	4.78
12.	16	108.49	77.40	4.45
13.	17	112.83	75.76	4.10
14.	18	117.07	74.25	3.85
15.	19	120.97	72.68	3.15
16.	20	124.50	71.06	2.81
17.	21	127.79	69.46	2.56
18.	22	130.28	67.60	2.20
19.	23	132.43	65.73	2.07
20.	24	134.35	63.90	1.84
21.	25	135.90	62.05	1.77
22.	26	137.36	60.31	1.49

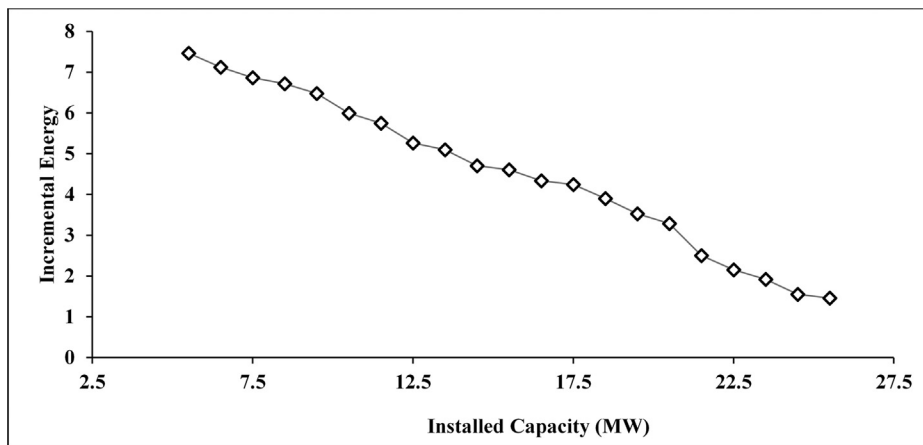


Figure 3.7 Graphical representation of installed capacity versus incremental energy.

3.7.2 Selection of unit-size and number of units

The steps for determining the size and quantity of producing units are as follows:

1. Minimum number of generating power units should be considered because the cost (generating unit's cost, supporting equipment cost, and civil works cost) increases with an increase in the number of units. Installation of a single unit has the drawback of complete power loss during a forced outage, which may be unacceptable in an off-grid system.
2. The number of units and their size will be determined by the machine's discharge requirements, allowing it to provide efficient/optimal power generation during lean discharge. If more than one unit is to be put in a power plant, they should all be of similar capacity in order to make inventory management easier. [Fig. 3.8](#) depicts the superposition of design discharges from an installation with equal unit rating on the stream's flow duration curve.
3. Unit size can be determined based on transportation constraints, such as the maximum size ($L \times W \times H$) of units that may be delivered to the project area.
4. There is a trend in certain situations to install varying capacity units to match the stream flow pattern; however, this led to varied unit sizes and an increase in inventory. If there is a significant increase of advantage in terms of improved energy output that may offset the increased inventory cost, then such a decision is preferable.
5. In the case of run-of-river systems without pondage, the number of units is determined by the varying discharge throughout the lean period, as well as the turbine operating parameters.

When the installed capacity is determined by monsoon season discharge and there is significant variance in lean/monsoon discharge, the number of units can be increased in comparison to projects with significant lean season flow.

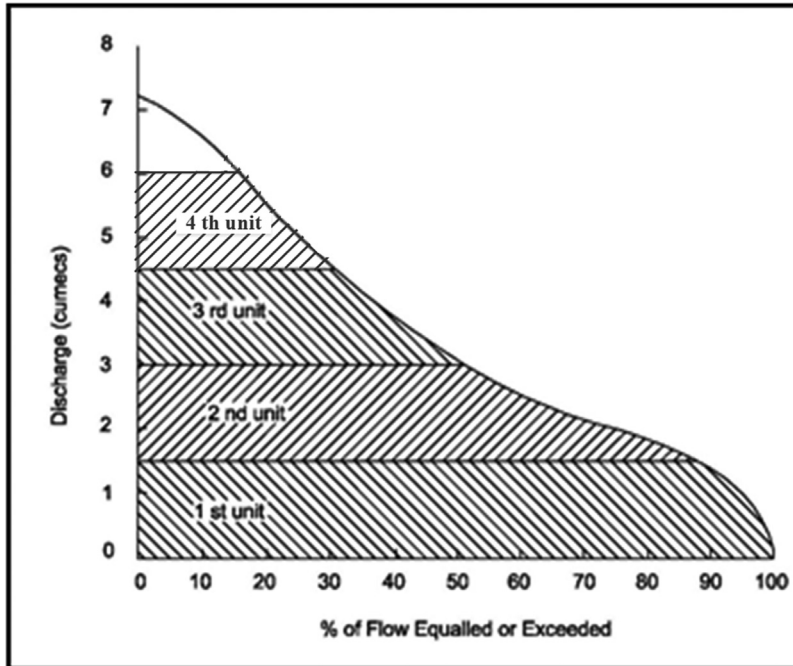


Figure 3.8 Equal sized turbine generating unit.

This will be determined by the machine's full/partial load requirements throughout the monsoon/lean season. This will also be influenced by the developer's prudent financial analysis criteria for electricity sales.

3.8 Summary

This chapter provides the range of site investigations need to be carried out for the development of SHP. The development of various works on proposed location is governed by various opportunities and limitations available on the existing topographical and geological features. Therefore site investigation is an important aspect in the preliminary and final stages of project development. The availability of water at the site governs the power potential and economic feasibility of the project.

Exercise

1. What is the difference between preliminary and detailed investigation?
2. What is the aim of site investigation for SHP projects?
3. Explain geological and topographical surveys required for different SHP schemes,
4. What is flow duration curve?

Table 3.3 Average monthly discharge data.

Average monthly discharge (cusecs)			
Month	Year 1999–2000	Year 2000–2001	Year 2001–2002
April	6748	7900	2948
May	7607	9176	6280
June	9876	10,352	10,232
July	2016	2997	7917
August	7646	6855	10,936
September	8895	8519	10,680
October	8652	4576	10,252
November	9450	8265	10,159
December	7248	7922	10,475
January	10,672	4193	7525
February	10,841	6565	1898
March	8760	7117	8419

5. How to determine hydropower potential? Write the steps required for fixation of installed capacity.
6. What are the parameters considered for selecting the number of units and their sizes?
7. Write in brief on:
 - a. Muck disposal
 - b. Load demand survey
 - c. Power evacuation survey
 - d. Difference between Survey of Grid connected and isolated systems
 - e. Socioeconomic survey
8. Work out the flow duration of the discharge data given in Table 3.3. Also, work out the installed capacity for 9.8 m head?

Further reading

- AHEC-IITR, “1.12 – Site investigations” standard/manual/guideline with support from Ministry of New and Renewable Energy, Roorkee, August 2013. https://hre.iitr.ac.in/departments/HRE/uploads/standards_pdf/1.12_Site_investigations.pdf.
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- S.K. Singal, R.P. Saini, C.S. Raghuvanshi, Analysis for cost estimation of low head run-of-river small hydropower schemes, Energy Sustain. Dev. 14 (2) (2010) 117–126. ISSN 0973-0826.
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